Some Extra Optimization Problems

1. What are the dimensions of the most economical cylindrical can having capacity of 100 cm^3?

① $V = \pi r^2 h$

$h = \frac{100}{\pi r^2}$

② $S.A. = 2\pi r^2 + 2\pi r h$

$= 2\pi r^2 + 2\pi r\left(\frac{100}{\pi r^2}\right)$

$= 2\pi r^2 + \frac{200}{r}$

③ $SA' = 4\pi r - \frac{200}{r^2}$

④ set $SA' = 0$

$4\pi r^3 - 200 = 0$

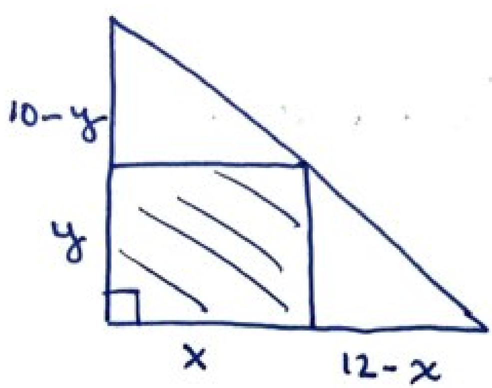
$r^3 = \frac{50}{\pi}$

$r = \sqrt[3]{\frac{50}{\pi}}$

$r \doteq 2.52\text{ cm}$

$h \doteq 5.03\text{ cm}$

2. Determine the area of the largest rectangle that can be inscribed in a right triangle with legs adjacent to the right angle having lengths of 10cm and 12cm. The two sides of the rectangle lie along the legs.



$A = x \cdot y$

$A = x\left(10 - \frac{5}{6}x\right)$

$A = 10x - \frac{5x^2}{6}$

$A' = 10 - \frac{5x}{3}$

set $A' = 0$

$10 - \frac{5x}{3} = 0$

$x = 6 \Rightarrow y = 5$

$\therefore x = 6\text{cm}$ & $y = 5\text{cm}$.

$\therefore$ Largest Area is 30 cm^2.

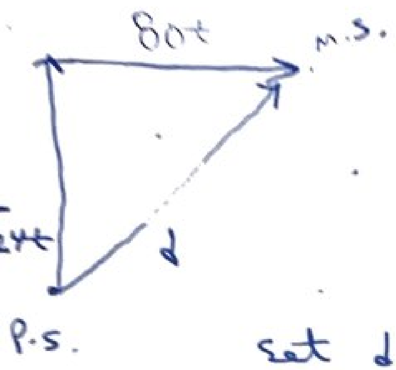
$\frac{10-y}{10} = \frac{x}{12}$

$x = \frac{6(10-y)}{5}$

$x = 12 - \frac{6}{5}y$

$y = 10 - \frac{5x}{6}$

3. At 1:00pm a merchant ship is sailing 18 km/hr due east. At the same time a patrol boat 40 km south of the merchant ship is travelling north at 24 km/hr. When will they be closest to each other?



$$s = \frac{d}{t} \Rightarrow d = st$$

$$d = \sqrt{(18t)^2 + (40-24t)^2}$$

$$d' = \frac{648t + 2(40-24t)(-24)}{2\sqrt{(18t)^2 + (40-24t)^2}}$$

set $d' = 0$

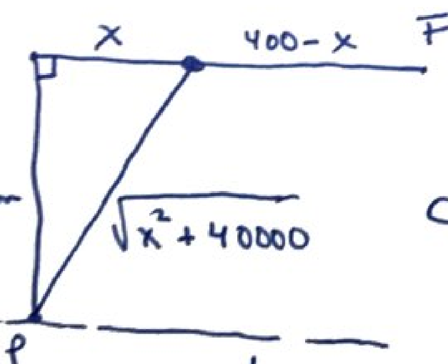
$$\Rightarrow 648t - 1920 + 1152t = 0$$

$$1800t - 1920 = 0$$

$$t = \frac{1920}{1800}$$

$t \doteq 1.067$ OR @ 2:04 pm.

4. A powerhouse, P is on one bank of a straight river 200 m wide and a factory, F is on the other side of the bank 400 m downstream from P. A power cable is suppose to connect P with F. The cost to lay the power cable underwater is \$12/m and the cost to lay the power cable underground is \$6/m. How should the power cable be laid to minimize cost?



$$C = \$12 \cdot \sqrt{x^2 + 40000} + \$6(400 - x)$$

$$c' = \frac{12x}{\sqrt{x^2 + 40000}} - 6$$

$$c' = 0 \Rightarrow \frac{12x}{\sqrt{x^2 + 40000}} = 6$$

$$2x = \sqrt{x^2 + 40000}$$

$$4x^2 = x^2 + 40000$$

$$3x^2 = 40000$$

$$x \doteq 115.47\text{m}.$$

∴ The cable should be placed on the bank at 284.53m from F.